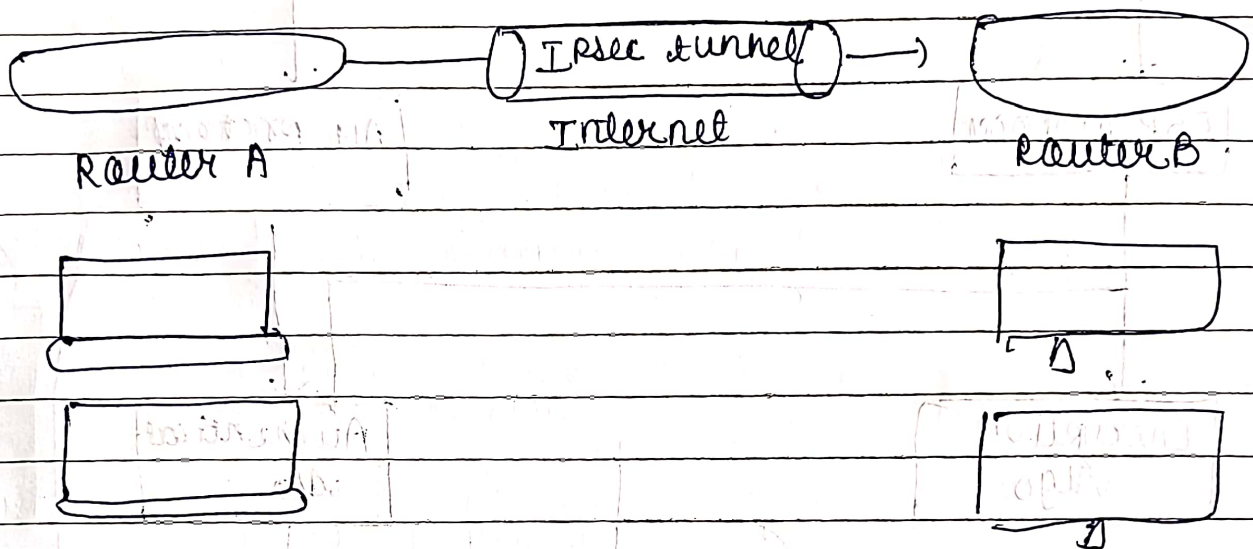


⇒ unit -4

⇒ 1) Security at different layer
[Application, Transport & N/w]

→ [Read]

→ IPsec Protocol



→ IPsec is a suite of protocols that provide n/w layer security for IP communication ensuring data confidentiality, integrity & authentication.

→ widely used in VPNs.

→ Key features

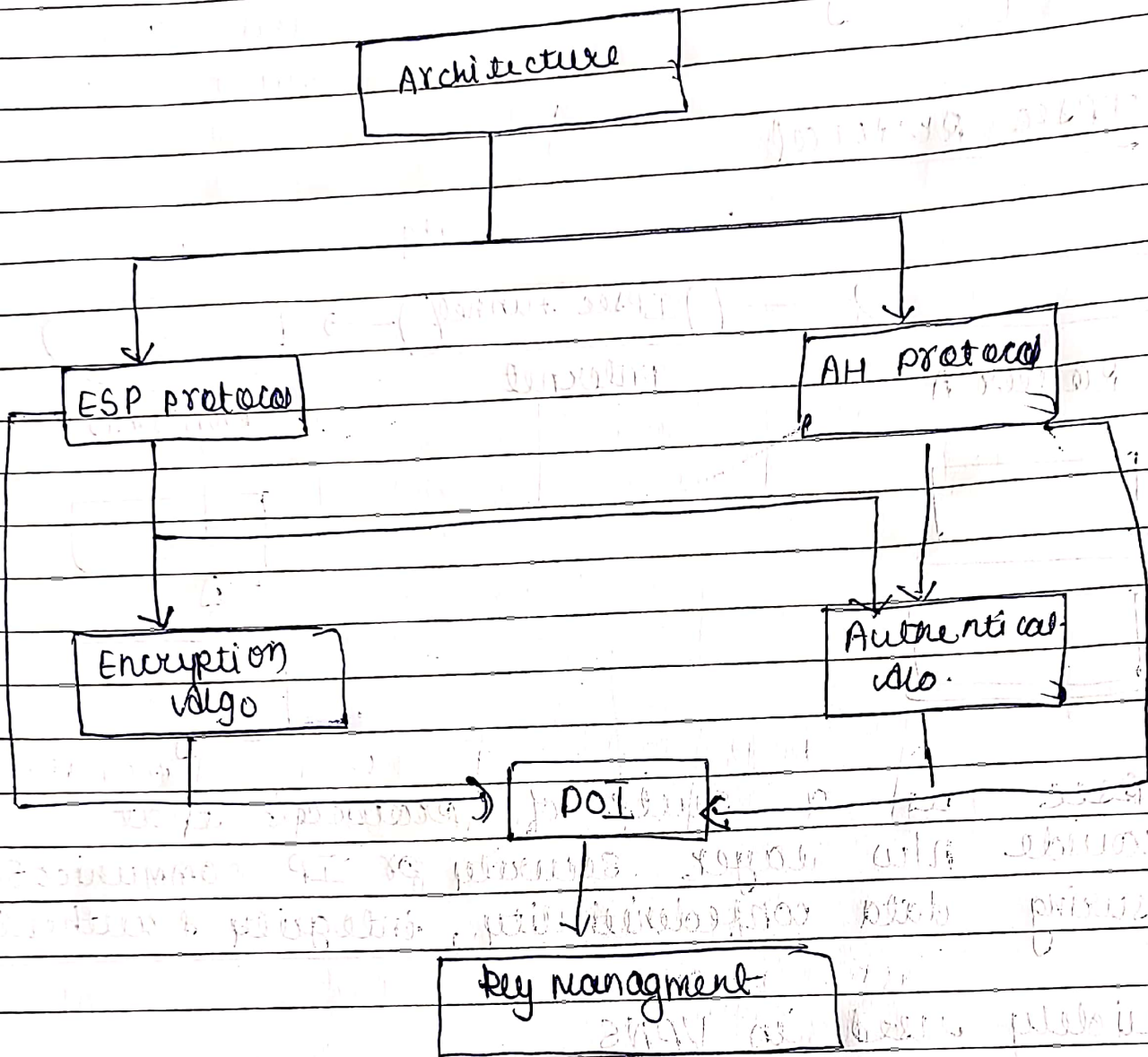
→ Encryption

→ Authentication

→ Data integrity

→ Anti Replay

→ IPsec Architecture



→ Explain this protocol in one-two line

→ Advantages.

→ It provides network-layer security. as it works on the n/w layer & provides transparency to application.

→ Provides confidentiality during any kind of data exchange.

→ It has zero dependability on application.

⇒ MIME

→ MIME stands for Internet mail Extension (MIME).

→ MIME is a kind of add on - or supplementary protocol that allows non-ASCII data to be sent through SMTP. It allows the users to exchange different kinds of data files on the internet, audio, image app.

→ Features

→ It is able to send multiple attachments with single message.

→ unlimited message length

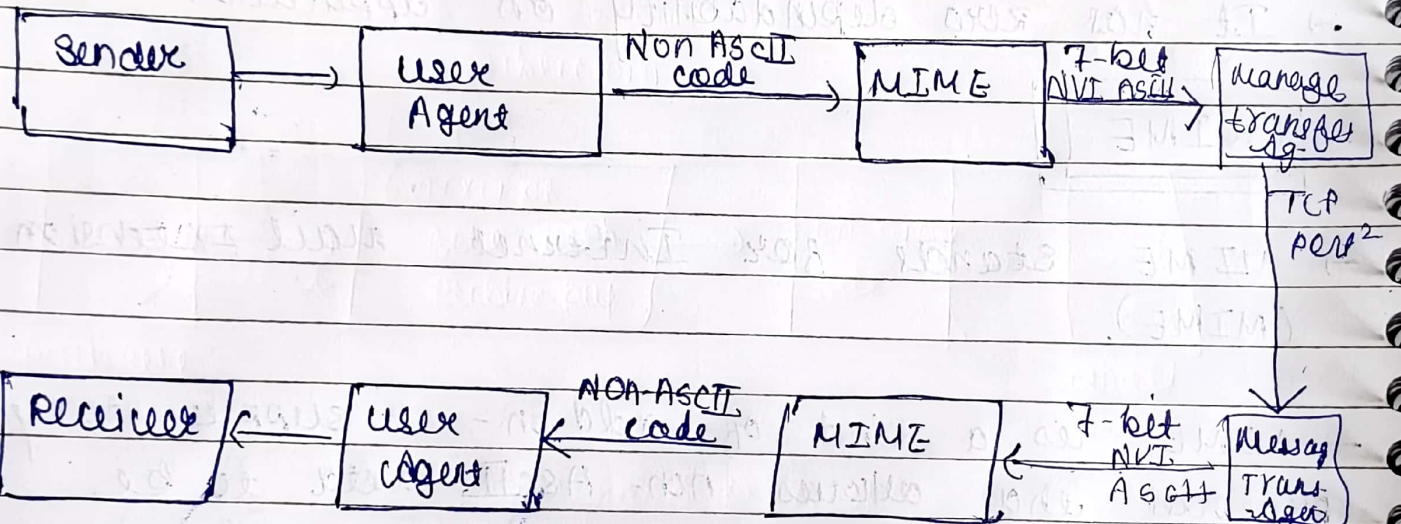
→ Binary attachments

→ Mime provided support for varying content types & multipart messages.

→ MIME Header

- MIME-version
- Content type
- Content type- Encoding
- Content-ID
- Content - Description

→



→ Include Advantages & Disadvantages

-1 PGP

→ Pretty Good Privacy is a widely used encryption and digital signature protocol for securing email and file communication.

→ It uses asymmetric encryption.

→ PGP also incorporates digital signatures to ~~verify~~ verify the authenticity and integrity of messages & file.

→ Key features

→ Asymmetric encryption

→ Digital signatures

→ Email security

→ File encryption

→ Hybrid approach

→ Open source.

→ How PGP works

→ Key Generation :- E

→ encryption

→ signing

→ Transmission

→ Decryption.

→ Benefits of using PGP.

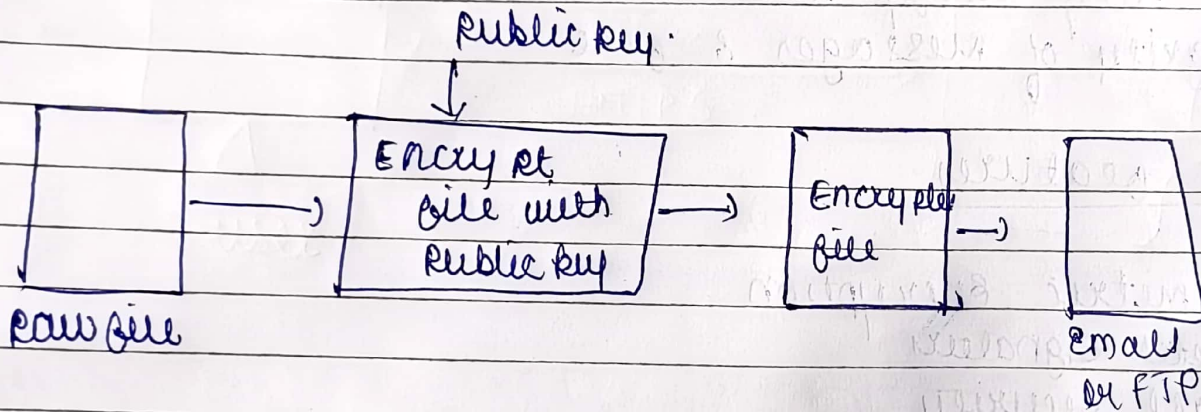
→ Confidentiality

→ Authenticity

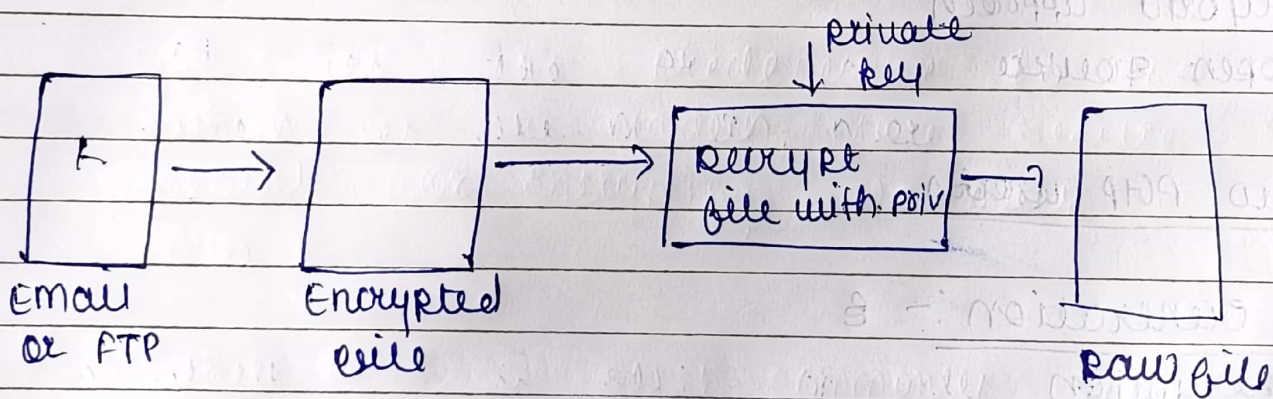
→ Integrity

→ Non Repudiation.

Note → Explain this terms according to marks.



Encryption process.



SSL OR TLS

Secure socket layer (SSL) is a security protocol that encrypts data transmitted over a n/w., like the internet to ensure privacy & integrity

⇒ key concepts

→ Encryption

→ data integrity

→ Authentication

→ Handshake.

→ How it works.

→ client Request → A web browser (client) sends a request to a web server.

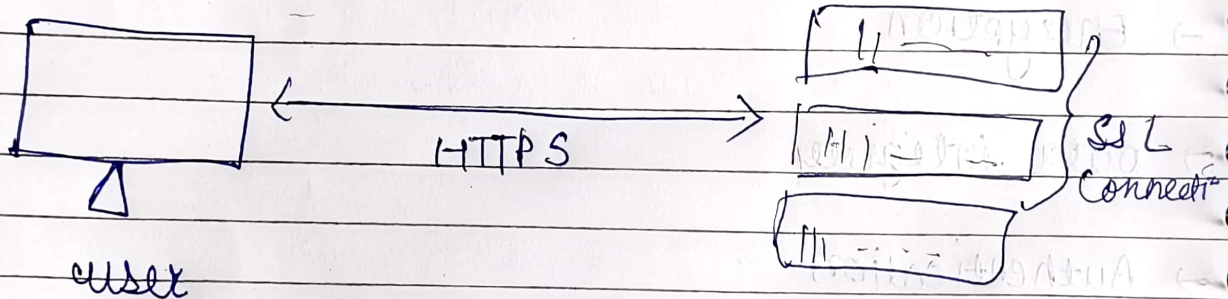
→ server response → The web server responds with its SSL certificate & other info.

→ SSL Handshake → The browser & server engage in handshake to establish a secure connection.

Encryption & Decryption.

→ Importance of SSL.

Security
Trust
compliance



→ TLS (Transport layer security).

SSL is the predecessor to TLS, which is the modern, more secure protocol for encrypting data over n/w.

→ while SSL is still commonly referred to, TLS is the current standard

→ where to find SSL.

HTTPS.

Padlock icon.